



Mark Scheme (Results)

January 2025

Pearson BTEC Nationals
In IT (31760H)
Unit 1: Information Technology Systems

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January 2025

Publications Code 31760H_2501_MS

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Unit 1: Information Technology Systems

General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- All marks on the mark scheme should be used appropriately.
- All marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if a candidate's response is not worthy of credit according to the mark scheme.
- Where some judgment is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt about applying the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed-out work should be marked UNLESS the candidate has replaced it with an alternative response.
- Phonetic spelling should be accepted.

Specific marking guidance

The marking grids have been designed to assess learner work holistically. Rows in the grids identify the assessment focus/outcome being targeted. When using a marking grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer, in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band, depending on how they have evidenced each of the descriptor bullet points.

BTEC Next Generation Mark Scheme Template

IT Unit 1 January 2025

Question Number	Answer	Mark
1a	Any two from: Mouse / trackerball / joystick (1) Stylus / pen (1) Digitiser (1) Scanner (1) Keyboard (1) Drawing/graphics tablet (1) Touch screen (1) Camera (1) Accept any other relevant input device.	2

Question Number	Answer	Mark
1bi	Award up to two marks for each of two linked explanations, from: • enables the designers to have a more flexible working pattern (1) because designers can access their files from home / not restricted to one workstation (1) • information is easily accessible (1) because all files are stored in one place (1) • designers can collaborate on a project (1) because they can share/access the same files (1) Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
1bii	Award up to two marks for each of two linked explanations, from: • improved data security (1) because it makes it easier to apply security measures / easier to monitor user activity (1) • easier to backup data (1) because the company can schedule automatic backups / because all files are stored in one place (1) • cost effective (1) because the company does not have to purchase/maintain additional equipment/storage (1) • scalability (1) because the company can update/expand the server when required (1) Accept any other relevant phrasing/wording Allow: examples of security measures	4

Question Number	Answer	Mark
1c	Award up to two marks for each of two linked explanations, from: • the connection will be secure/encrypted (1) prevents unauthorised access to the device (1) • improved privacy when using shared computers (1) as all traffic is kept inside the secure VPN 'tunnel' (1) • avoids bandwidth throttling (1) because it hides the activity from their (Internet Service Provider) ISP (1)	4

Question Number	Indicative content
1d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the features of suitable image file formats. <u>JPEG</u> • lossy compression • suitable for sharing images • universal browser and OS support • fairly low file size • lossy image compression might lead to poor text readability • supports 24-bit colour <u>GIF</u> • lossless compression • limited palette of colours (limits to 8 bits per pixel) • animation support • low file size • 8-bit limitation leads to limited image quality • supported by all major browsers and OS and image editors <u>PNG</u> • lossless compression • maintains detail and contrast between colours • high quality • larger files can slow down the website if overused (especially high-resolution images) • supported by all major browsers and OS • transports transparency <u>TIFF</u> • supported by most digital platforms and devices • commonly used for storing and editing images • typically used as a lossless image format • high quality files perfect for storing or printing publishing • large file size due to no compression • limited browser support without add-ons or file extensions • mainly used as an export format for professional image editing and publishing tools <u>BMP</u> • lossless • maps individual pixels with no compression • can become extremely large file sizes but retains high quality and remain the same when you upload them • supported by all major browsers • most default OS image editor

	<u>SVG</u> • 2-dimensional vector images • web-friendly format, can be read by search engines using keywords • can be resized without losing quality • small file sizes (XML) • can be edited with text editors/drawing software • supports animation and interactivity
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Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Answer	Mark
2ai	Award one mark for identification and two additional marks for appropriate expansions. • File management software (1) is used to help browse, search, and organise files on the system quickly (1) • Disk cleanup tool/uninstaller (1) is used to free up disk space by removing unnecessary/temporary files (1) • Backup utility (1) is used to create a copy of stored data / can automatically back up your data / to ensure lost data/files can be retrieved (1) • File compression tools (1) are used to reduce the size of files for storage / files that are sent as email attachments (1) • Disk defragmenter (1) is used to rearrange fragmented data into contiguous blocks to reduce read/write times (1) • Firewall software (1) because it monitors incoming and outgoing traffic / blocks unauthorised access (1) • VPN software (1) because it encrypts the network connection / sends data packets through a secure tunnel (1) • Encryption software (1) so that data is unreadable / can only be understood by authorised users (1) Accept any other relevant phrasing/wording Do Not Accept: 'scrambled' for encryption in this instance.	4

Question Number	Answer	Mark
2aii	A description of how the college's antivirus software will protect the network. To include any four from: • (Anti-virus software) scans computer/files (1) • compares threats detected against a database of definitions (1) • analyses the code/behaviour of the suspicious program (1) • determines the nature of the threat (1) • alerts the user (1) • give option to / will remove/quarantine the threat (1) • log user actions to deal with similar future threats (1)	4

Question Number	Answer	Mark
2b	Award up to two marks for each of two linked explanations, from: • An ID card system does not require the staff to enter a code (1) therefore unauthorised users / shoulder surfers could not copy the number (1) • A four-digit code can easily be leaked to (numerous people) (1) whereas the ID card can only be passed to one person (at a time) (1) • the college can (quickly) deactivate the card (1) because the member of staff would report their card has been lost or stolen promptly (notice it has gone) (1) • ID cards cannot be copied (easily) (1) because they require specialist equipment / have in built 'technology' (1) • An ID card is unique to the card holder (1) to ensure that the user of the card is the person authorised (to access the server room) (1) Accept any other relevant phrasing/wording Accept mix and match if the response makes sense.	4

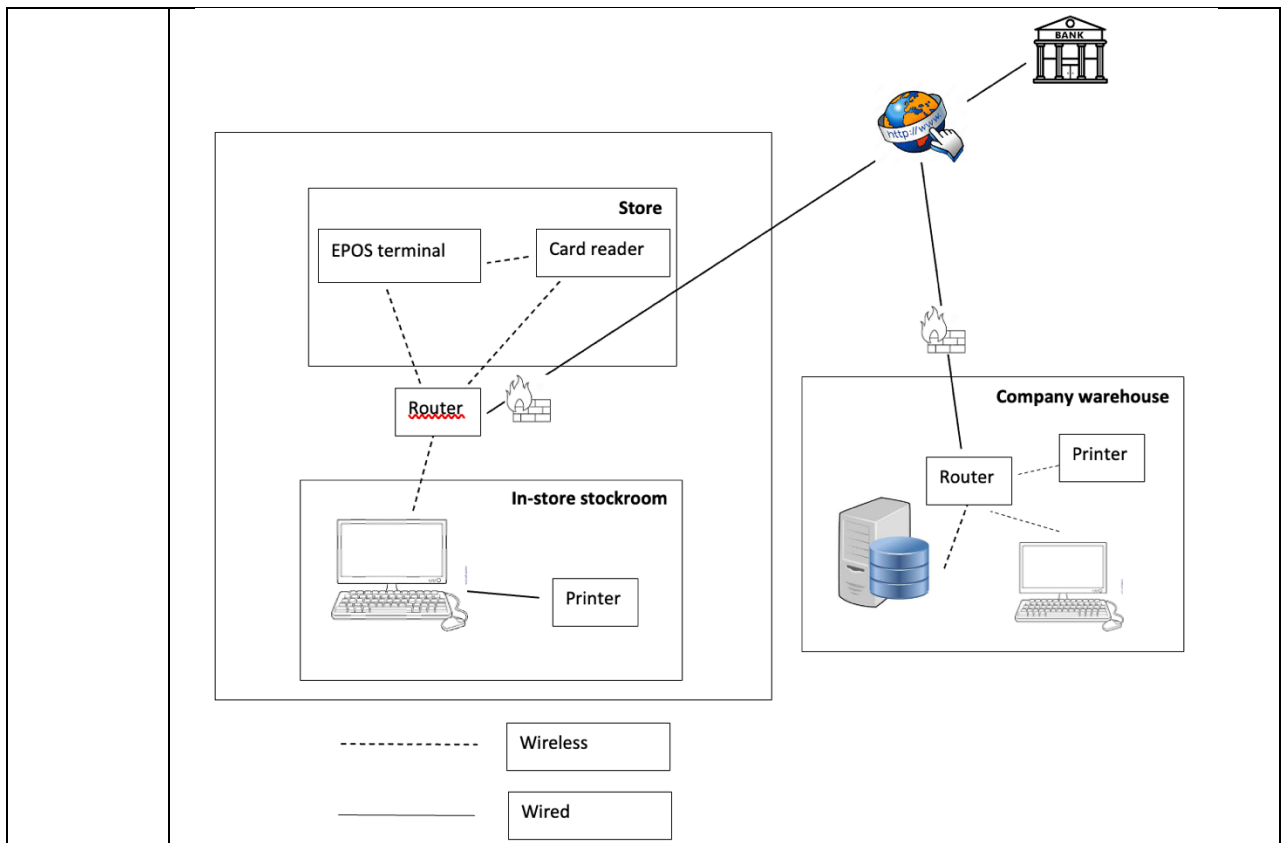
Question Number	Answer	Mark
2c	Award one mark for each point up to a maximum of four marks. • input student's username and password (1) • system checks username and password against a database of saved usernames/passwords (1) • grant access if a match (1) • display error message if no match / asked to re-enter details (1) • system allows a limited number of attempts (1) • account is locked when unsuccessful attempts exceed number of attempts (1) Accept any other relevant phrasing/wording Allow: 'email' for 'username'	4

Question Number	Indicative content
2d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the moral and ethical factors for staff and students when using IT in the classroom. <u>Privacy</u> • where portable devices such as phones are used and what is being said to avoid eavesdropping/passing on of private/sensitive data • who can see computer screen - switch off monitor/put machine into sleep mode when not using/ moved away from PC • who can see contents of documents/emails • make sure passwords conform to college standards / are not written down / not shared with other staff / students • privacy policies of websites. <u>Online behaviour and netiquette</u> • treating others with respect / use appropriate language / do not engage in cyber bullying • be professional when dealing with others • only visit trusted websites • be aware of malware and phishing – do not open links / attachments in emails from unknown senders • do not upload / post illegal / offensive materials that could cause offence to others • breach copyright laws when downloading materials • respect the privacy of others • consider tone when posting comments • do not shout using capitals <u>Acceptable Use</u> • internet use should be restricted to research for college purposes • file permissions should be enforced so staff and students do not access other people's folders without permission • IT systems (internet, email and mobile technologies) should only be used for college purposes • downloading / installing software onto college systems should be prohibited • take care using IT equipment at all times – do not pull any cables out / bypass web filters • do not access social media websites unless authorised • 'think before you print' – only print out what is essential, saving both paper and energy • switch devices off or into sleep mode when not in use • set power saving features • ensure that 'redundant' materials are correctly recycled/disposed of e.g. printer cartridges, can they be refilled

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4-6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7-8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
3a	A diagram of a potential system/integration of IT systems and connections used to meet the requirements of the scenario. Example response – note this is indicative only. Credit should be given for a diagram that meets the requirements of the given problem. Diagram may include: Devices: • Stock room – Desktop PC, and printer • Store – EPOS terminal, card reader • Warehouse – database server, desktop PC, printer, switch, and router Connection types and users: EPOS terminals • EPOS terminal to desktop PC – wireless / wired • EPOS terminal to internet – wired • Card reader to customer’s bank – wired Store stock room • Desktop PC to EPOS system – wireless / wired • Desktop PC to warehouse – wired Company warehouse • Internet to database server – wired • Internet to desktop PC – wired • Desktop PC to database server – wired/wireless • Desktop PC to printer – wired/wireless Annotations: The diagram should include annotations for devices and systems as well as the connection types used.



Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-2	- Diagram provides partial coverage of devices. - Limited connections are shown - Diagram includes some annotations of the connection types used.
Level 2	3-4	- Diagram provides coverage of mostly appropriate devices. - Connections are shown which meet most requirements of the scenario. - Diagram includes some annotations of connection types used to meet the majority of the requirements of the scenario, but these may not always be the most efficient.
Level 3	5-6	- Diagram provides thorough coverage of appropriate devices which fully meet the requirements of the scenario. - Connections are shown which fully meet the requirements of the scenario. - Diagram includes detailed and accurate annotations of devices and connection types used to meet the requirements of the scenario.

Question Number	Answer	Mark
3b	Award up to two marks for each of two linked explanations, from: • file permissions (1) because these control who can access specified files and what actions they are permitted to perform (1) • access levels (1) allows users to access specific data and systems (as defined in their job role) / designates the type of access (read only or read/write) • backup and recovery procedures (1) is the process of duplicating data and storing it in a secure place in case of loss or damage so it can be restored at a later date (1) • passwords (1) are used to allow authorised staff access to data and systems (1) • physical access control / biometrics (1) are used to determine which staff are authorised to enter areas of the warehouse (1) • anti-virus software (1) which scans for viruses (and alert the user to either remove or quarantine the threats) (1) • firewalls (1) which monitors incoming and outgoing traffic / blocks anything suspicious (1) • encryption (1) this protects data and systems by scrambling data so it is unreadable (1) • VPN (1) creates a secure tunnel that protects data when sending to external sources (1) Accept any other relevant phrasing/wording	4

Question Number	Answer	Mark
3c	Award up to two marks for each of two linked explanations, from: • to verify identity of website / verify that a company owns a website (1) which proves that the information is genuine/valid/legitimate (1) • protect data integrity (1) to ensure that data has not been tampered with (1) • verifies that the website is authentic/legitimate/official (1) so that antivirus software will allow access to the site (1) • indicates the website can be trusted/reassures customers (1) allowing customers to know their personal/payment details are safe (1) • digital certificates encrypt the communication (between the web browser and website) (1) to protect customers' personal/payment details (1) Do Not Accept – answers relating to showing website is secure eg Padlocks or https Accept any other relevant phrasing/wording	4

Question Number	Indicative content
3d	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners will discuss the security protocol websites use for data transmission. <u>HTTPs (Hypertext Transfer Protocol secure)</u> • is a secure version of HTTP • encrypt users HTTP requests and webpages to increase security of data transfer • protects against eavesdroppers/man in the middle attacks <u>SSL (Secure Socket Layer)</u> • Provides privacy and data integrity for internet communications • Encrypts the link between a website and browser / two servers • Ensures data transmitted between a client and server remains private • Prevents hackers from reading any data transmitted • The client is able to authenticate the identity of the server <u>TLS (Transport Layer Secure)</u> • Provides privacy and data integrity for internet communication • More secure version of SSL • Ensures the secure delivery of data over the internet / avoids eavesdropping • Can be used for other applications such as emails / file transfers / video conferencing / VOIP

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	4–6	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	7–8	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
4a	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners discuss the impact on the company and its staff of moving to cloud storage. Positive impacts <u>Company</u> • Cost: no need to pay for technical staff, either in-house or on a consultancy basis, to provide a backup system this will be done automatically. • Storage capacity can be scaled up and down to fit the needs of the business. This has the added benefit of being better for the environment than running a server with excess capacity. • Will give Clothes for Us robust disaster recovery, which would otherwise be unlikely due to lack of finance and expertise. • Reduces the implications/security risks of lost or stolen laptops and other mobile devices. Data stored on the cloud is automatically backed up and therefore instantly retrievable. • Improved security arises from the reduction in the need to send files via external methods such as email. <u>Staff</u> • Because the system can be used anywhere with an internet connection, staff will be able to access the files/folders when they are out of the office or working remotely. • Many suppliers provide mobile apps that will allow staff to access data on a range of devices. • Some staff, e.g. office staff, may be able to take advantage of the system and work from home. • Because all files are stored centrally everyone sees the latest/same version leading to improved collaboration. Reducing the problems of conflicting file content, formats and titles. • Can have their own logins and access their files from their own personal storage area Negative impacts <u>Company</u> • Introducing a third party may have security implications for Clothes for Us if the 3rd party does not have robust security procedures in places for backup and recovery and data transfer • Training of staff would be time consuming and costly to the company and could result in downtime while training is taking place • Cost: ongoing subscription payments to the 3rd party provider, expensive if large amounts of storage is required for the designs • Downtime: designers could not complete designs in time, which could result in other companies having a competitive advantage

	<u>Staff</u> • Must have an internet connection, could be a disadvantage to staff out of the office. • Impact of downtime – staff will not be able to access their files if the cloud storage is down or no internet, which could lead to deadlines being missed and payment could be withheld • Staff might not want all files stored on a shared area because other people could copy their designs and use for their own purposes • Frustrating for staff to get used to new technology which can affect their work productivity
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Mark scheme (award up to 10 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited discussion which contains generic points • Little or no consideration of different aspects
Level 2	5–7	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed discussion which considers some different aspects • There will be some consideration of how they interrelate
Level 3	8–10	• Demonstrates mostly accurate and detailed knowledge and understanding • There will be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical discussion which clearly considers a range of different aspects • Clearly shows how they interrelate

Question Number	Indicative content
4b	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Learners evaluate the features of the network that designers could use to work collaboratively. Shared area: • Can be set up to centrally store designs so designers can access the shared area remotely • Allows all designers to access/edit documents from a central storage point • Version control can be maintained to ensure designers are working on the most current version • Applications running on the network can enable designers to work in real time Management of groups: • Only authorised members of the organisation will have access to the share area • Designers will have access to the same software to create their designs • Ensure no compatibility issues with the designers using devices remotely

Mark scheme (award up to 12 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-4	• Demonstrates isolated elements of knowledge and understanding • There will be major gaps or omissions • Few of the points made will be relevant to the context in the question • Limited evaluation which contains generic or descriptive points • Little or no conclusion
Level 2	5-8	• Demonstrates some accurate knowledge and understanding • There will be some gaps or omissions • Some of the points made will be relevant to the context in the question, but the link will not always be clear • Displays a partially developed evaluation which considers some different competing points • There may be a conclusion which is partially supported
Level 3	9-12	• Demonstrates mostly accurate and detailed knowledge and understanding • There may be minor gaps or omissions • Most of the points made will be relevant to the context in the question, and there will be clear links • Displays a well-developed and logical evaluation which clearly considers different aspects and competing points in detail • There is a conclusion that is supported

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